Figure A.1 Actor, use case, and stimulus

An *actor* is someone or something that interacts with the system you are designing. A use case provides some value to the actor. Let’s analyze a problem and design the use case diagram for it to clarify this process.

> **PROBLEM STATEMENT** Design a vending machine: a customer inserts coins and chooses a drink that is delivered. For simplicity, consider only the case with no exceptions and with the exact amount of coins required deposited.

The use case diagram for the vending machine coin counter and drink dispenser is shown in figure A.2. It has four use cases at the first level--insert coins, see drinks, select drink, and pick up drink--which are direct stimuli or operations invoked by the customer. The insert coins operation in turn invokes the count coins use case, see drinks requires display drinks, and select drink results in deliver drink. These secondary use cases are not directly invoked or used by the customer. An important note is that a use case diagram is not like a traditional flow chart. It simply lists the operations in the elliptical use case symbols. The operational flow is not defined here.